Project status report

**Status key**

- Complete
- On track
- In trouble
- Needs immediate attention

## Executive overview:

**Project name/ID:** PlanER — Team 07 (COMP 8157, Fall 2026)

**Project manager:** Shashang Patel & Nauman Syed

**Date:** November 3, 2026

**Project summary:** PlanER compiles a declarative entity-resolution request (two collections, a budget or F1 target) into the cheapest MongoDB plan: a blocking index, a cascade of classical and LLM matchers, and escalation thresholds. Midterm: storage, all four blocking indexes and the full matcher cascade are built, and a fixed-plan prototype runs end to end.

## Overall project health:

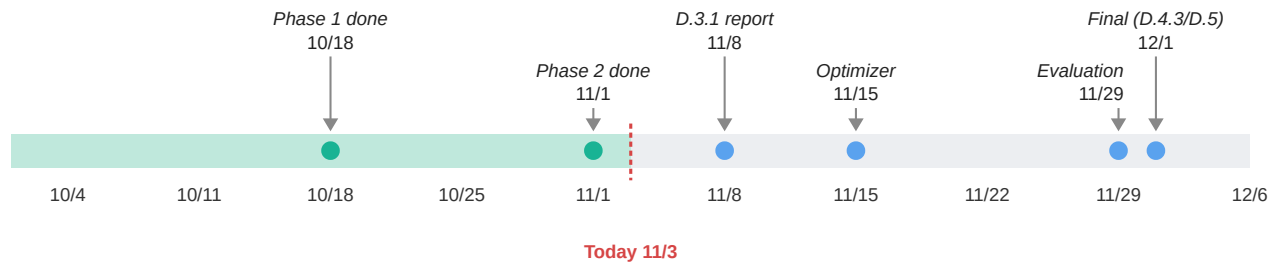
***On track***

**Notes:** Phases 1 and 2 finished on schedule (Oct 18, Nov 1). Phase 3, the cost-based optimizer, started Nov 2 as planned. 12 of 20 TBS tasks completed, 2 in progress, 6 remaining; no slippage against the D.2.1 timeline.

## Project health breakdown:

| Element | Status | Owner | Notes |
|---|---|---|---|
| Environment & data ingestion | ***Complete*** | S. Patel | T1–T2: Docker stack; 4 benchmarks loaded with tokens, keys, embeddings |
| Blocking access paths | ***Complete*** | S. Patel | T3–T7: B-tree, multikey, $search, $vectorSearch; profiled with explain() |
| Classical matchers | ***Complete*** | N. Syed | T8–T10: rules, Fellegi–Sunter (EM), random forest |
| LLM tier & verdict cache | ***Complete*** | N. Syed | T11: Ollama (Llama 3.1 8B, Qwen 2.5 7B); llm_cache reuse |
| Cost-based optimizer | ***On track*** | N. Syed / S. Patel | T13–T16: profiling and estimation in progress; due Nov 15 |
| Repair, evaluation & demo | ***On track*** | Both | T17–T20: scheduled for Phases 4–5 (Nov 16 – Dec 1) |

## Milestone review:




### Risks:

**Known issues:**

- $search / $vectorSearch (mongot) are public preview and may be unstable.
  - **Action plan:** all access paths sit behind one interface; vector path can switch to the Atlas free tier.
- 7–8B local LLMs are slow on laptop hardware.
  - **Action plan:** only escalated pairs reach the LLM; every verdict is cached in llm_cache.

**Potential risks:**

- Optimizer cost/F1 estimates may be poor when matches are rare in the sample.
  - **Action plan:** stratified sampling; vary sample size and report estimation error.

**Change request:**

- Replace pgvector/FAISS (pre-proposal) with MongoDB $vectorSearch.
  - **Action plan:** approved in D.2.1; implemented in T6.

### Team progress:

**Team's progress:**

- On schedule. S. Patel led storage and blocking; N. Syed led matching and the LLM tier; both review each other's pull requests.

**Tasks completed:**

- Docker stack and data ingestion (T1–T2)
- Four blocking access paths and profiling (T3–T7)
- Rules, Fellegi–Sunter, random forest (T8–T10)
- Local LLM tier with verdict cache (T11)
- Fixed-plan end-to-end prototype + Streamlit (T12)

**Tasks planned for next period:**

- Matcher profiling and plan estimation (T13–T14)
- Plan enumeration with Pareto pruning (T15)
- Optimizer-driven runs: predicted vs observed (T16)
- Transitivity repair (T17), then full evaluation (T19)